DBA 802 final verification report

Course: DBA 802 — Data Analytics & Strategic Decision Intelligence
Student name supplied for the course: Divya. Student ID: not supplied.
Manuscript author: Divya Munagala.
Study: A knowledge-guided early-warning system for digital-transformation failure in entrepreneurial firms (KG-EWS).
Target journal: Knowledge-Based Systems (Elsevier, ScienceDirect).

Recommendation: MINOR REVISIONS. The research files in this package match the finalized manuscript. Requirement 6 is not met, because there is no genuine submission confirmation. The package is not ready for professor submission until that screenshot is real.

1. What was checked

The manuscript PDF and Word file in folder 01 are byte copies of manuscript.pdf and manuscript.docx. The cohort table has 120 primary rows and 60 stress rows. Means in results.json round to the three-decimal figures in Tables 1 and 2, including AUC 0.765, cost 0.598, and the logistic cost advantage 0.024 with counts 59 / 28 / 33. Figure 1 is the plotted paired-cost figure, 1950 × 1350 pixels. No model was refit for this package.

The stored illustrative case is a failed, rule-triggered firm with evidence 3.40 and rules R1, R2, R3, R7. Rounded probabilities are 0.118 (logistic), 0.016 (gradient boosting), 0.971 (knowledge base), and 0.999 (KG-EWS), matching the manuscript.

2. References

The manuscript cites 29 sources, numbered in order of appearance. All 29 DOIs were matched to Crossref on 24 October 2026 for year, volume, and author family names. Quartile categories were read on SCImago for 2024 and 2025 on that same date and are transcribed in Reference_Quartile_Record.xlsx.

Twenty-two of those sources are counted as Q1 and are published by ScienceDirect (17), IEEE (3), or MDPI (2). Mathematical Biosciences is Q2 and is not counted. Six further sources sit outside those three publishers. Their quartiles are not asserted. Knowledge-Based Systems itself is Q1 in the recorded 2024 and 2025 categories. Journal of Theoretical and Applied Electronic Commerce Research is Q1 in Business, and its Computer Science Applications category is Q2 in 2025.

The Wilcoxon entry keeps pages 80–83. The Crossref page field is 80 only. That bibliographic difference is unresolved.

3. Figures and tables

One figure and two tables are in the manuscript, and the same three items are in folder 02. There is no second figure. The graphical abstract is a separate drawing of the same method and the same three-decimal results. It does not introduce a performance number that the manuscript does not report.

4. Argument the manuscript actually makes

KG-EWS adds eight committee rules to a logistic log-odds. The evidence weight and the alert threshold minimize a decision cost that charges a missed failure five times a false alarm. Across 120 synthetic cohorts the combined score has the lowest mean cost and the highest mean AUC of the four instruments. The gain over logistic regression is small, paired, and absent when the evidence weight is zero or when the predicates are flipped. The rules alone catch rule-triggered failures by alarming on 92.2% of surviving firms.

The business claim in the manuscript is a decision claim: the warning that is issued is the combined score

together with the names of the rules that fired, and it is cheaper than the logistic score only under the stated cost and a correct coding of the rules. There is no separate heading called managerial implications. That material is in sections 4, 5, and 6.

5. Professor requirements

No.	Requirement	Status
1	Complete research paper	Complete and verified. Byte copy of the journal manuscript. Contact details and student ID are absent.
2	Tables, images, and graphs	Complete and verified for the one figure and two tables that exist.
3	Supporting data	Complete and verified. Data are synthetic and generated by ews.py.
4	Graphical abstract	Complete and verified against the manuscript numbers and the Elsevier minimum pixel size.
5	Supplementary materials	Complete and verified. Configuration, quartile record, highlights, and competing-interests file.
6	Submission screenshot	Awaiting actual journal submission. No confirmation exists.

6. Issues that remain

- No genuine Elsevier submitted-status screenshot.
- Affiliation, full postal address, and corresponding-author e-mail are not in the manuscript.
- Student ID was blank in the assignment instructions.
- The package filename uses Hemasri_Babburu. The author line is Divya Munagala. The author line was not changed.
- The PDF reference list is compact so that 29 sources fit the 10-page short-communication limit. The body is double-spaced.
- Wilcoxon pages 80–83 versus a Crossref page field of 80.
- Six bibliography sources were not given a SCImago quartile in this audit.
- The study is a synthetic recovery benchmark. Rule thresholds match the generator. There is no failure date. The five-to-one cost is a stated preference. Predicate noise removes the advantage over logistic regression.

No new findings were generated for this package. No submission screenshot was drawn.